Electric Charge
Electrons have negative charges protons positive
Q = charge
SI Unit = Coulomb (C)

Voltage is potential energy
SI Unit: Volt
Symbol: V

Ground is relative

Current is flow of charge
rate at which electrons "travel"
SI Unit: Ampere (Coulomb per second)
Symbol: A

Kirchoff's Current Law

Total current flowing into an area is equal to the total current flowing out of that area

Resistance

Measure of how much material resists the flow of current
SI Unit: Ohm (Ω)
Symbol: R

Material property

Ohm's Law

The voltage across a resistor equals the current through the resistor times its resistance

$V = IR$

Voltage = Current x Resistance

Ground breaking

Follow me!

Power

Instantaneous Rate at which energy is transferred

Power equals voltage across times current through material

$P = IV$

Power = Current · Voltage

$P = \frac{V^2}{R}$

Power = $\frac{\text{Voltage Squared}}{\text{Resistance}}$

Substitution for current using Ohm's Law

Symbol = P
SI Unit = Watt (W)

Energy

The ability to make a change (do work)
Symbol: E
SI Unit: Joule (J) = one watt second

Also common, Watt Hour which is equal to 3600 watt seconds

Capacitance

A measure of how well an object can store charge / resist change in voltage
Measured in charge per voltage
Symbol: C
SI unit: Farad (F)

Analogy: a water reservoir
Voltage is the depth of the water, capacitance is the area of the reservoir

How much an object resists change in voltage
Analogy: the reservoir takes time to fill or drain

Inductance

A measure of how well an object can resist a change in current

Stores energy in the "momentum" of the current flowing through it
Symbol: L
SI Unit: Henry H

Analogy:
Heavy turbine in a river; Like a big water wheel
you already know this

Up we go!

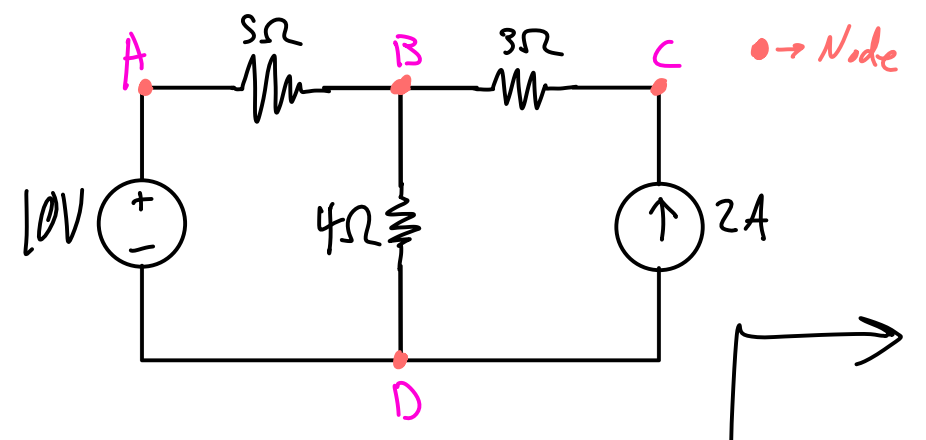
Circuits!

A complete continuous loop that allows current to flow all the way back to where it came from.

Made up of nodes and components (or devices) between these nodes

A NODE is a "wire" where voltage is the same everywhere across it

Actually useful ↗
Always confused me in ECE-80T

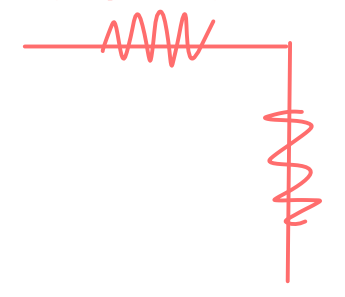


Series and Parallel

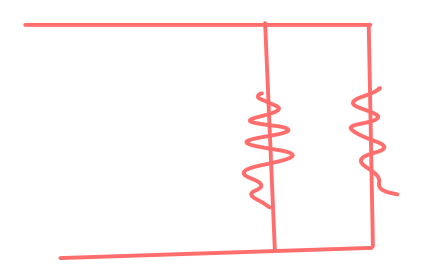
A series circuit is one where devices are placed one after another

A parallel circuit is parallel

Series



Parallel



Resistors

Resist stuff yo

Has power rating indicating the maximum amount of power a resistor can dissipate

Have to take into account the environment they're in as well

Thermistors and Potentiometers

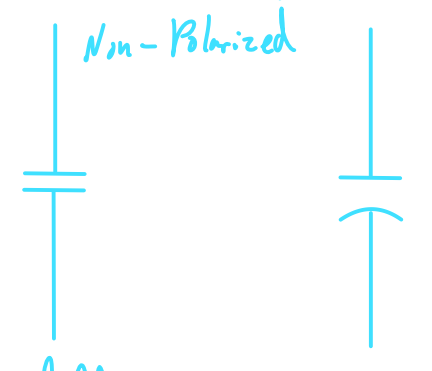
Thermistors: change their resistance depending on temperatures and what's
◦ non-linear behavior Approximated by third degree polynomial

Potentiometer: tunable resistance
Why do we use them?

Still going!

Capacitors

A component designed to contain a specific capacitance
Often have a maximum voltage rating
Has memory of past voltage — can store energy
Made from many conductive plates between them



difference between polarized and non-polarized:
Polarized capacitors are assymetrical
voltage travels one way

Inductors

A component designed to contain a specific inductance
Maximum Current
Pass to much current they'll not

Saturation Current

How much current the coil can take before they stop acting like inductors due to a magnetic field breakdown
Has alot of
Ferrite cores increase inductance

Wires

Wires carry current
more current creates more heat (see power section)
Narrower wires have higher resistance per length → more heat
Wires are measured in two ways
AWG
american wire gauge
mm²
square milimeters
cross sectional area

can be stranded or solid core
can be shielded

Fuses

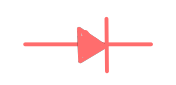
Fuses "protect" "down stream" circuits
blows up before other stuff blows up

Relays (and Contactors)

Electromechanical switches
Use a magnet to open or close a circuit
Have voltage and current ratings (look at data sheets)
Has lots of convoluted terminology like SPST or DPST
Provide isolation / High voltage protection

Diodes

A device which restricts current to only allow it to flow one way
Make from chemically doped silicon
Have a cathode side and an anode side
Consistent

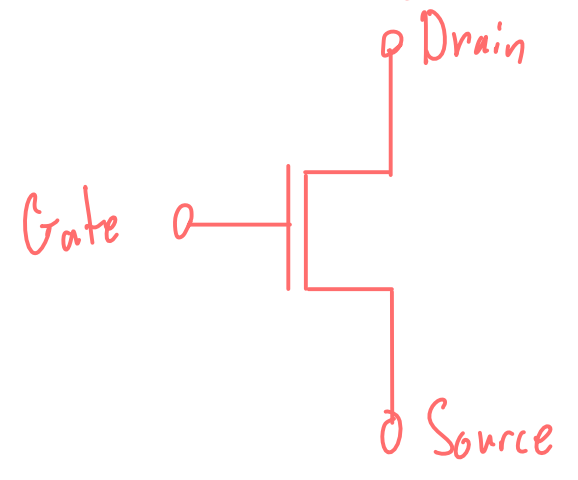


Transistors

transistors are switches (like relays) except they have no moving parts
Made from semi conductors

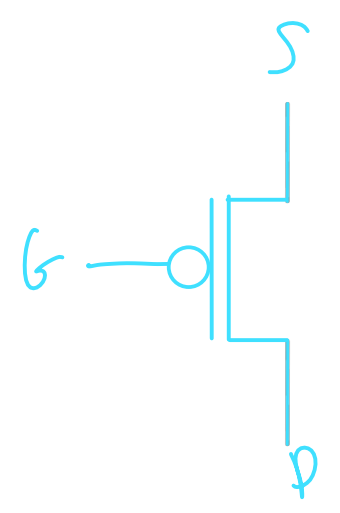
N Channel MOSFET

Transistor that uses the field Effect to switch
- Allows current to flow from the drain to the source when the gate voltage is a certain threshold greater than the source voltage
- Usually source is connected to ground
- Really good at pulling circuits down



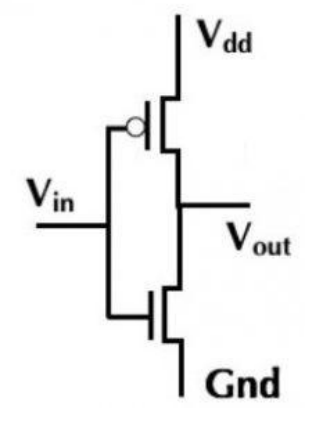
P Channel Mosfet

Similar to NMos but current flows from source to the drain instead of the other way around



CMOS (complimentary MOSFet)

- NMOS can pull circuits down to a lower voltage while
- PMOS can pull up to a higher voltage
- These can be combined to efficiently set an output to a 1 or a 0



Bipolar Junction Transistor BJT

BJT's are another type of transistor
NPN and PNP types
Current driven instead of voltage driven
Voltage drop instead of voltage driven
V

Operational Amplifier

Its output voltage is the difference of its inputs times a really big number (ideally infinite, though not in practice)

Allows you to amplify really small signals

Luca/Jack cover them in office hours

Comparators

integrated circuit that gives a digital output depending on its analog inputs

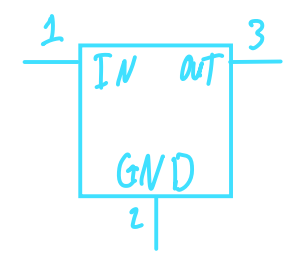
Hysteresis and Schmitt Triggers

Skipped for a reason

produces a pulse whenever a magnetic field passes by

Linear Regulators

Integrated circuits that steps a Voltage down to a fixed (sometimes adjustable) voltage
Has maximum current rating / power rating



Holy smokes dude
long lecture